

PortelX

Universal Temporary Identity Layer — Software Requirements Specification & Phased Roadmap

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Primary Domain: Ephemeral Identity / UPI	Institution: SBCET / RTU Kota	Classification: Confidential

CORE VALUE PROPOSITION: "Borrow. Do. Disappear."

PortelX establishes an ephemeral, verifiable cryptographic micro-session on an arbitrary foreign Host Device that enables critical financial transactions, sovereign document inspection, and corporate single sign-on. Upon lifecycle completion or exit trigger, the runtime executes sub-second memory shredding (< 1,000ms), ensuring an absolute zero-residue digital footprint.

1. Executive Summary & Problem Analysis

PortelX is an ephemeral identity runtime layer that enables an **Initiator** to securely project their digital identity—including payment capabilities, sovereign credentials, health records, corporate authorizations, and digital signatures—onto an arbitrary foreign **Host Device** inside a cryptographically bounded, isolated execution bubble.

The Problem: Modern digital authentication rigidly binds a user's operational capabilities (UPI payment apps, corporate VPN credentials, DigiLocker storage) to a single physical device. When that hardware is incapacitated (battery exhaustion, physical damage, device theft, or confiscation during transit), users face an unbridgeable dilemma: forfeit critical access or endure high-risk credentials leakage on foreign/borrowed devices. Logging into someone else's device persists authentication cookies, cached credentials, OTPs, SMS notifications, and persistent local storage files.

The PortelX Solution: PortelX completely decouples operational identity from static hardware. It replaces persistent logins with temporary, isolated micro-sessions verified through Multi-Factor Biometric Attestation and zero-trust orchestration. As soon as the user completes their transaction, a sub-second cryptographic shredding routine purges all volatile memory buffers, encryption keys, and cache partitions, guaranteeing an absolute **zero-residue footprint**.

Strategic Architectural Differentiation: Access Conduit vs. Storage Vault

Operational Dimension	Traditional Cloud Wallets & DigiLocker	PortelX Ephemeral Runtime Layer
Data Storage Model	Persistent file vaults. Downloads documents, PDFs, and authentication cookies directly to local flash storage.	Pure access conduit. Streams cryptographically verified proofs into RAM capsules; zero bytes touch NAND flash.
Foreign Device Residue	High risk: Session cookies, cached tokens, and download histories persist indefinitely after closing.	Zero residue: Sub-second cryptographic zeroization (<1s) completely shreds memory upon session end.
Device Dependency	Rigidly bound to SIM cards, primary hardware IDs, and physical phone storage.	Hardware-agnostic projection: runs securely across verified host smartphones, tablets, or kiosks.

2. System Architecture & Zero-Trust Mechanics

PortelX operates on a high-throughput, three-tier zero-trust topology designed for sub-second lifecycle control and hardware-backed biometric verification.

- **Client Tier:** Consists of two decoupled native applications: (1) *Host Device Runtime* (React Native) providing an isolated container with application-layer sandboxing, and (2) *Primary Authenticator* running on the user's primary hardware (or secondary trusted device) managing WebAuthn passkeys and Firebase Cloud Messaging (FCM).
- **Edge & API Tier:** Cloudflare API Gateway & WAF providing DDoS shielding and SSL termination, routing to high-concurrency Python 3.11 FastAPI asynchronous microservices. Backed by an in-memory Redis Ephemeral Cache running sub-second TTL key eviction.
- **Persistence & Verification Services:** Supabase (PostgreSQL 15+ with Row-Level Security) managing salted, pseudonymized audit logs; OpenAI Risk Engine calculating real-time behavioral anomaly scores; and RevenueCat managing dynamic subscription entitlements.

2.1 Multi-Factor Biometric Attestation (MFA Engine):

Before any temporary session capsule can be initialized on a host device, PortelX enforces strict **100% convergence across three distinct cryptographic factors** (zero partial privilege elevation permitted):

1. **Live Host Facial Biometrics:** Real-time facial geometry attestation captured via host camera hardware and matched against encrypted vector proofs.
 2. **Native Host Hardware Fingerprint:** Cryptographic biometric attestation queried through Android BiometricPrompt (Class 3) or iOS LocalAuthentication.
 3. **Out-of-Band Primary Passkey Challenge:** High-entropy challenge payload pushed via WebAuthn/FIDO2 to the initiator's primary Secure Enclave.
- *Progressive Security Enforcement:* If sequential factor validation fails 3 consecutive times, the backend engages a progressive 300-second lockout on the user account and flags the host device fingerprint.

2.2 Ephemeral Session Lifecycle & Sub-Second Memory Shredding:

Upon successful 3-factor convergence, the backend injects an AES-256-GCM encrypted session capsule into the host runtime. The host maintains continuous vitality verification through full-duplex WebSockets. A strict, non-extendable 600-second inactivity countdown timer is enforced.

Immediate Failsafe Termination: The temporary container self-destructs instantly upon: (a) transaction completion, (b) manual 'Terminate' tap, (c) WebSocket heartbeat disconnect, (d) OS window blur / app backgrounding, (e) OS device locking, or (f) timeout expiration.

Cryptographic Zeroization Routine: Within **< 1,000 milliseconds** of termination trigger, the client runtime overwrites all heap buffers with random bytes, zero-fills memory, flushes cache partitions, invalidates the session token on Redis, and securely shreds all ephemeral AES keys in memory.

3. Runtime Boundaries & Autonomous Fraud Protection

Allowed Inside Session (Whitelisted)	Blocked by Design (Blacklisted Subsystems)
• NPCI-compliant UPI payment intent construction. • Real-time camera BharatQR / UPI QR scanning. • Read-only account balance querying. • Ephemeral in-memory inspection of sovereign documents. • Execution of cryptographically signed digital receipts.	• Filesystem: Access to host photos, contacts, and storage. • Clipboard: Native clipboard inspection and copy-paste blocked. • Screen Capture: Screenshots and recording blocked via FLAG_SECURE (Android) and view masking (iOS). • IPC & Sockets: Inter-process communication and foreign outbound network sockets blocked.

Real-time Autonomous Fraud Dispatch:

Within **5,000 milliseconds** of session creation, the backend pushes an out-of-band high-priority alert (FCM / APNs) to the user's primary registered hardware with: *Host Hardware Fingerprint, Approximate Geolocation, Counterparty, Amount, and Unix Timestamp*. The user selects '**Recognized**' or '**Dispute & Terminate**'. Disputes immediately freeze the session and initiate an investigative quarantine pipeline cross-referencing biometric attestation telemetry and audit logs.

4. Technology Stack & Subscription Tiers

The PortelX technology stack is purpose-built for maximum concurrency, cryptographic integrity, and cross-platform native execution.

Architecture Layer	Technology Selection	Technical & Architectural Rationale
Frontend Mobile Runtime	React Native (Expo / Bare Workflow)	Rapid cross-platform native compilation for Android and iOS devices.
Backend API Engine	FastAPI (Python 3.11 Asynchronous)	High-concurrency async I/O, native ML compatibility, auto-generated OpenAPI.
Primary Database & RLS	Supabase (PostgreSQL 15+)	Instant edge data APIs, built-in Row-Level Security, real-time WebSocket subscriptions.
Ephemeral State Store	Redis Cloud (Sub-second TTL)	Sub-millisecond read/write latency with auto-expiring keys for session token invalidation.
Authentication Standard	WebAuthn / FIDO2 + Platform Biometrics	Hardware-backed asymmetric public-key cryptography; zero stored passwords.
Dynamic Risk Scoring	OpenAI API (GPT-4o Mini) / Isolation Forest	Low-latency behavioral risk evaluation and device anomaly classification.
Entitlement & Billing	RevenueCat Mobile SDK	Cross-platform receipt validation and dynamic tier feature gating.
Out-of-Band Push Rail	Google Firebase Cloud Messaging (FCM)	Sub-5-second high-priority delivery to initiator's primary smartphone.

Commercial Monetization Architecture (RevenueCat Gating):

Feature Dimension	Free Tier	Premium Tier	Enterprise Tier
Monthly Sessions	5 Ephemeral Sessions	Unlimited Sessions	Uncapped / Policy Governed
AI Risk Scoring	Standard Rule Checks	Real-Time Behavioral AI	Advanced Threat Modeling
Trusted Circle	Disabled	Up to 5 Family Members	Multi-Tenant Org Control
Audit Log Cloud Sync	Local Memory Only	30-Day Encrypted Retention	Infinite SIEM Log Export
Enterprise MDM Link	Disabled	Disabled	Full Zero-Trust MDM Link

Commercial Enterprise Security Rationale:

While individual consumers utilize PortelX for personal emergency liquidity and travel verification, corporate enterprise accounts unlock high-margin recurring SaaS revenues. Enterprise plans empower corporate CISOs to govern contractor and travelling auditor access on external client workstations, enforcing ephemeral zero-trust boundaries with full SIEM audit compliance and zero residual corporate secrets.

5. Phases of Project — Complete Roadmap to 100% Achievement

To achieve **100% full production readiness, enterprise adoption, regulatory compliance, and global scalability**, PortelX is structured into six comprehensive, sequential execution phases. Each phase defines explicit technical deliverables, security validation benchmarks, and quantifiable percentage progress milestones.

Master Phased Implementation Matrix:

Phase	Phase Title & Strategic Focus	Timeline	Target Progress	Core Milestone Deliverables
Phase 1	Research, Cryptography & Architecture	Weeks 1–3	20%	Ephemeral protocol math, threat modeling, memory zeroization benchmarks, WebAuthn spec.
Phase 2	Hackathon MVP & Proof of Concept	Weeks 4–8	40%	Dual React Native apps, FastAPI backend, Supabase RLS, mock UPI, OpenAI risk scoring, FCM alerts.
Phase 3	Alpha Hardening & Biometric Bridges	Months 3–4	60%	Native BiometricPrompt/LocalAuth bridge, WebSocket heartbeat, sub-second shredding, VAPT audit.
Phase 4	Beta Launch, Live Banking & Docs	Months 5–7	80%	NPCI-certified UPI bridge, DigiLocker memory streaming, Trusted Circle P2P, DPDPA 2023 compliance.
Phase 5	Enterprise Suite & MDM Profiles	Months 8–10	95%	Corporate SAML/OIDC SSO, OEM/MDM profiles, SIEM log sync, Kubernetes scaling to 10k sessions.
Phase 6	100% Achievement: Global Ecosystem	Months 11–12+	100% COMPLETE	W3C DID cross-border identity, transit/kiosk SDK, ABHA healthcare, SOC2 Type II, 99.99% SLA.

Phase 1: Research, Cryptographic Architecture & Core Specification (0% → 20%)

Strategic Objective: Formulate the underlying cryptographic protocol, model all potential mobile attack surfaces, and prove the mathematical feasibility of volatile-only execution without persistent NAND flash storage.

- **Cryptographic Protocol Formulation:** Define the non-reversible User Identity Token (UIT) generation algorithm and ephemeral AES-256-GCM session key derivation using cryptographically secure random number generators (CSPRNG).
- **Volatile Memory Zeroization Benchmarks:** Prototype low-level C and native mobile wrappers to execute memory buffer overwrites (random byte overwrite followed by zero-fill) to guarantee deallocation within 1,000ms.
- **Threat Modeling & OWASP Compliance:** Conduct formal threat modeling against OWASP Mobile Security Testing Guide (MSTG Level 2), documenting analog hole boundaries, screen recording vectors, and IPC leak paths.
- **WebAuthn & FIDO2 Architecture:** Formulate challenge-response schemas binding primary user authentication to hardware Secure Enclaves (Apple Secure Enclave and Android Titan/StrongBox MIDs).

Phase 2: Hackathon MVP & Proof of Concept (20% → 40%)

Strategic Objective: Deliver a functional, interactive end-to-end prototype demonstrated on physical mobile devices, showcasing core temporary session lifecycles, mock payment execution, and instant memory shredding.

- **Dual Client Applications:** Build the *Host Runtime App* and *Primary Authenticator App* using React Native (Expo bare workflow) with unified UI design and seamless QR scanning.
- **Asynchronous API Engine:** Deploy high-concurrency FastAPI microservices integrated with Redis Cloud for ephemeral token lifecycle management and automatic TTL eviction.
- **Database & Security Rules:** Deploy Supabase PostgreSQL with strict Row Level Security (RLS), storing salted user profiles and anonymized, pseudonymized transaction audit logs.
- **Simulated Financial Rails:** Build UPI deep-link intent routing and an in-memory mock settlement ledger simulating real-time payment clearance and receipt generation.
- **AI Anomaly Demonstration:** Connect OpenAI API (GPT-4o Mini) to score session context (host fingerprint, time, location delta) and flag anomalous logins.
- **Commercial Entitlement & Alerts:** Integrate RevenueCat SDK for tier feature gating and Firebase Cloud Messaging (FCM) for real-time primary phone push alerts.

Phase 3: Alpha Testing, Native Hardware Attestation & Security Hardening (40% → 60%)

Strategic Objective: Transition from mock and application-layer wrappers to native hardware-enforced biometrics, robust session state synchronization, and rigorous penetration testing.

- **Native Hardware Biometrics Bridge:** Replace simulated authentication with native bridges to Android BiometricPrompt (Class 3 Strong Biometric hardware) and iOS LocalAuthentication / FaceID.
- **Full-Duplex Vitality Heartbeat:** Deploy WebSocket edge routing that continuously validates session vitality; any socket disruption, network loss, or app backgrounding triggers immediate local shredding.
- **Progressive Security Rate Limiting:** Implement progressive backend lockouts (300 seconds following 3 consecutive factor failures) to eliminate brute-force attack vectors.
- **Automated Dispute & Quarantine Engine:** Build the server-side investigative pipeline that freezes tokens and cross-references attestation logs when a user taps 'Dispute & Terminate'.
- **Vulnerability Assessment & Pen Testing (VAPT):** Engage external ethical security researchers for comprehensive SAST, DAST, and reverse-engineering testing (Frida hooking, jailbreak bypass resistance).

Phase 4: Beta Launch, Banking Gateway Partnerships & Sovereign Documents (60% → 80%)

Strategic Objective: Replace simulated financial flows with certified NPCI banking rails, integrate national document repositories, and conduct closed beta user trials.

- **Production UPI & Payment Gateway Clearing:** Establish banking sponsorship and integrate certified Payment Aggregator (PA) APIs for real-time live UPI settlement with NPCI compliance.
- **DigiLocker & Sovereign Document Streaming:** Connect with DigiLocker and health repositories to allow read-only document inspection directly in memory capsules, strictly prohibiting local file caching.
- **Trusted Circle Peer-to-Peer Network:** Implement cryptographic peer-to-peer invitation and verification, enabling users to designate trusted family, friends, or corporate colleagues as verified host nodes.
- **Closed User Group (CUG) Public Beta:** Roll out production beta builds to 1,000+ selected users across diverse mobile hardware, capturing telemetry, latency metrics, and real-world failure cases.
- **Regulatory Privacy Compliance:** Perform formal legal and technical audit under the India Digital Personal Data Protection Act (DPDPA 2023) and GDPR, enabling automated 'Right to Erasure' API endpoints.

Phase 5: Enterprise Fleet Management & Mobile Device Management (MDM) (80% → 95%)

Strategic Objective: Unlock enterprise B2B monetization by integrating enterprise identity providers (IdPs), Mobile Device Management (MDM) profiles, and organizational compliance consoles.

- **Corporate Identity Federation (SSO):** Support enterprise SAML 2.0 and OIDC identity federation, allowing travelling executives and contractors to launch temporary single sign-on corporate sessions.
- **OEM & Enterprise MDM Sandbox Profiles:** Partner with MDM systems (Android Enterprise, Apple Knox, Jamf) to enforce kernel-level display capture negation and operating system clipboard isolation.
- **Centralized Enterprise Administration Console:** Deliver a web dashboard for enterprise sysadmins to monitor active fleet sessions, configure device whitelists, review compliance digests, and trigger instant remote revocations.
- **Enterprise SIEM Log Export:** Automate real-time streaming of encrypted, pseudonymized security event logs into enterprise SIEM platforms (Splunk, Datadog, AWS CloudWatch, IBM QRadar).
- **Cloud Infrastructure Horizontal Scaling:** Containerize backend microservices on Kubernetes across multi-region Cloudflare edge clusters, stress-tested to sustain 10,000+ concurrent active sessions.

Phase 6: 100% Achievement — Global Production, Hardware Enclaves & Ecosystem Scale (95% → 100%)

Strategic Objective: Complete the strategic evolution into a globally recognized temporary identity standard, expanding into international cross-border identity capsules, public transit/kiosk SDKs, and critical healthcare.

- **W3C Decentralized Identifiers (DID) & Verifiable Credentials:** Implement W3C-compliant self-sovereign identity standards for cross-border digital passports, travel visas, and international credentials.
- **Hardware FIDO2 Security Keys Support:** Support physical security keys (YubiKey, Google Titan) via NFC and USB-C for offline and zero-network passkey attestation.
- **Public Kiosk & Transit Terminal SDK:** Deliver hardened kiosk runtime binaries for airport self-service check-in, high-speed train transit gates, and hospitality self-check-in kiosks.
- **Emergency Healthcare Dossier Access:** Integrate Ayushman Bharat Health Account (ABHA) and international health standards for emergency medical personnel access during critical accidents.
- **Formal Industry Certifications:** Secure formal SOC 2 Type II certification, ISO/IEC 27001 accreditation, and CERT-In empanelled audit sign-off.
- **High Availability SLA:** Formalize 99.99% core infrastructure uptime availability backed by 24/7 global follow-the-sun enterprise engineering support.

6. Project Governance, Revision History & Author Affiliation

PortelX is governed under rigorous engineering management protocols ensuring architectural lineage, traceability of non-functional constraints, and cryptographic auditability across all execution milestones.

Governance Parameter	Specification Details
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